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MindTrace: Text Mining and NLP-Driven Emotion Prediction Using Machine Learning and Deep Learning Approaches

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Abstract

Text-based emotion detection from social media has grown in importance as platforms such as Twitter generate large volumes of emotionally expressive informal text. This study proposes MindTrace, a comparative evaluation of four classification models — SVM, XGBoost, CNN, and BiLSTM — applied to a 416,809-sample Twitter corpus annotated across six emotion categories: anger, joy, sadness, fear, surprise, and love. Existing comparative studies vary widely in dataset selection, preprocessing methods, and evaluation metrics, making direct performance comparisons unreliable. This study addresses that gap by fixing all experimental conditions across all four model families on the same corpus. A standardised NLP preprocessing pipeline handles tokenisation, stopword removal with negation preservation, lemmatisation, and class balancing. ML models use TF-IDF vectorisation; DL models use trainable embedding layers. Performance is assessed using macro-F1, weighted-F1, and class-wise confusion matrices to account for significant class imbalance (imbalance ratio $\approx 9.5{:}1$*). The findings are expected to provide empirically grounded guidance for architecture selection in emotion classification tasks, with applications in social media analytics, mental health monitoring, and emotion-aware AI systems.*

Keywords: emotion classification, natural language processing, machine learning, deep learning, BiLSTM, CNN, SVM, XGBoost, TF-IDF, Twitter, multi-class classification, social media analytics

1. Introduction

1.1. Background and Motivation

Emotion detection from textual data has become increasingly significant in the digital era, where individuals frequently express their psychological states and reactions through social media platforms. Automated emotion recognition systems enable deeper understanding of user behaviour, going beyond traditional sentiment polarity analysis. Such systems have practical applications in marketing analytics, customer relationship management, healthcare monitoring, and human-computer interaction (Nandwani and Verma, 2021; Poria et al., 2019).

Unlike sentiment analysis, which categorises content as positive, negative, or neutral, emotion detection aims to identify discrete emotional states such as anger, joy, sadness, fear, surprise, and love. However, accurately classifying emotions remains challenging due to contextual ambiguity, informal language usage, sarcasm, and class imbalance within datasets (Seyeditabari et al., 2018). This proposed study intends to develop a structured NLP pipeline and conduct a comparative evaluation between traditional machine learning and deep learning approaches for multi-class emotion classification.

1.2. Problem Statement

Despite significant research progress, a fundamental methodological gap persists. Comparative analyses across studies frequently differ in dataset selection, preprocessing strategies, feature engineering approaches, and evaluation metrics — making observed performance differences attributable to experimental inconsistencies rather than intrinsic model capabilities. Multi-class emotion classification across semantically similar categories (*e.g.*, fear vs. surprise, sadness vs. love) remains an unsolved challenge, further compounded by class imbalance in real-world social media datasets.

1.3. Research Question

How do traditional ML models (SVM, XGBoost) compare with deep learning models (CNN, BiLSTM) in multi-class emotion classification from social media text, when eval-

uated under a unified preprocessing pipeline, standardised dataset partitioning, and consistent evaluation metrics?

1.4. Research Objectives

The primary objective of this study is to develop a structured NLP-based framework for multi-class emotion prediction from social media text and to conduct a controlled comparison of traditional machine learning and deep learning models under identical experimental conditions. Specifically, the study:

1. examines the impact of preprocessing strategies, feature representation, and class balancing on model performance;
2. analyses class-wise prediction behaviour across models; and
3. provides empirical insights into the effectiveness and suitability of different modelling approaches for emotion classification.

2. Literature Review

2.1. State of the Art

Emotion detection research has progressed from lexicon-based methods to machine learning and, more recently, deep learning architectures. Early approaches used predefined resources such as the NRC Emotion Lexicon and WordNet-Affect to map words to emotion categories, but they were limited in handling context, figurative language, and emerging slang in social media text (Mohammad and Turney, 2013; Strapparava and Mihalcea, 2008).

Traditional machine learning methods improved performance through feature engineering using Bag-of-Words and TF-IDF representations, with models such as SVM and Naïve Bayes performing well on structured datasets, though they remained limited in capturing sequential and long-range context (Dzikovska et al., 2012; Wang, 2018). Deep learning further improved emotion classification by learning contextual patterns directly from text. CNNs were effective in extracting local n-gram features, while LSTM and BiLSTM models captured sequential dependencies more effectively (Hasan et al., 2019; Sun et al., 2023). XGBoost also provided a strong non-linear baseline for feature-based text classification (Chen and Guestrin, 2016).

More recently, transformer-based models such as BERT have achieved state-of-the-art performance by learning rich contextual embeddings, although their computational cost remains a practical limitation (Acheampong et al., 2021; Devlin et al., 2019). Across these approaches, class imbalance and inconsistent evaluation remain persistent challenges in emotion detection research (Nandwani and Verma, 2021).

2.2. Related Work on Base Models

This study considers four representative model families frequently applied in text classification:

Support Vector Machine (SVM) is a margin-based classifier that performs well in sparse, high-dimensional feature spaces such as TF-IDF representations but cannot capture sequential word relationships (Dzikovska et al., 2012).

XGBoost is a regularised gradient-boosted decision tree ensemble that improves predictive performance through iterative boosting and L1/L2 regularisation, though it also relies on engineered features rather than sequential text modelling (Chen and Guestrin, 2016).

Convolutional Neural Networks (CNN) extract local n-gram features through convolution and pooling operations (Hasan et al., 2019).

Bidirectional LSTM (BiLSTM) captures long-range contextual dependencies by processing token sequences in both forward and backward directions (Sun et al., 2023). Prior studies have shown that BiLSTM often outperforms classical baselines on multi-class emotion datasets due to its ability to model contextual semantics.

2.3. Prior Work Comparison

Table 1 summarises representative studies on text-based emotion classification, highlighting the dataset, model type, best reported macro-F1, and key experimental limitations.

Existing studies vary widely in dataset selection, preprocessing methods, evaluation metrics, and model choices, making direct performance comparisons difficult. Among the studies surveyed, no prior work has evaluated SVM, XGBoost, CNN, and BiLSTM on the same 416,809-sample Twitter corpus under identical preprocessing, stratified splits, and macro-F1 evaluation. Although transformer models such as BERT represent the current state of the art (Acheampong et al., 2021; Devlin et al., 2019), they are excluded due to computational constraints. This study instead focuses on a controlled comparison of classical and sequence-based architectures, with transformer-based extensions proposed as future work.

3. Methodology

3.1. Data Source

This study utilises the publicly available Emotion Dataset from Kaggle (Nelgiriye Withana, nd), a large-scale collection of Twitter posts annotated into six discrete emotion categories: anger, joy, sadness, fear, surprise, and love. The dataset comprises 416,809 labelled text samples, making it statistically robust for supervised multi-class classification modelling.

The dataset is methodologically justified on three grounds. First, its large sample size enhances statistical reliability and reduces variance in model estimation. Second, it reflects

Table 1. Summary of representative prior studies — dataset, models, performance, and experimental gaps.

Study	Dataset	Models	Best F1	Key Limitation
Saravia et al. (2018)	Twitter (6 cls, ∼20K)	CNN + word emb.	0.79 macro	Small dataset; single model family
Hasan et al. (2019)	Twitter multi-class	CNN, LSTM	∼0.82 acc.	No class-balancing; accuracy only
Demszky et al. (2020)	GoEmotions — Reddit (27 cls, 58K)	BERT fine-tuned	0.46 macro	Different domain; 27 classes
Sun et al. (2023)	Twitter sentiment	BiLSTM	∼0.85 acc.	Accuracy-only; inconsistent preproc.
Acheampong et al. (2021)	Multiple benchmarks	BERT (survey)	N/A	Survey only; no unified comparison
Proposed Study	**Kaggle Twitter (6 cls, 416K)**	**SVM, XGBoost, CNN, BiLSTM**	**TBD**	**Unified pipeline; macro-F1 eval.**

Table 2. Data dictionary — Kaggle Emotions Dataset.

Column	dtype	Role	Description
`index`	int	Index	Row identifier
`text`	object	Input	Raw Twitter post text
`label`	int64	Target	Encoded emotion class

Table 3. Sample records from the Kaggle Emotions Dataset.

Idx	Text (sample content)	Label
0	i feel really helpless and heavy hearted	4 (Fear)
1	ive enjoyed being able to slouch about...	0 (Sadness)
2	i gave up my internship...feeling lost	4 (Fear)
3	i dont know i feel so lost	0 (Sadness)
4	i am a kindergarten teacher...dedicated	4 (Fear)

informal, real-world social media language — including abbreviations, emotive expressions, and conversational tone — thereby increasing ecological validity. Third, its six-class taxonomy enables fine-grained emotional analysis beyond binary sentiment polarity, directly aligning with the research objective of multi-class emotion prediction.

Each instance consists of a raw textual input (`text`) and a corresponding encoded emotion label (`label`). Table 2 presents the complete data dictionary.

The label encoding follows: 0 = Sadness, 1 = Joy, 2 = Love, 3 = Anger, 4 = Fear, 5 = Surprise. Table 3 presents five representative records drawn from the dataset.

3.2. Exploratory Data Analysis

The objective of EDA is fourfold: (1) characterise class distribution and quantify imbalance to inform balancing strategies; (2) examine text length and word count distributions to guide feature engineering and sequence padding decisions; (3) investigate correlations among numerical features to assess redundancy; and (4) identify preprocessing requirements arising directly from observed data characteristics.

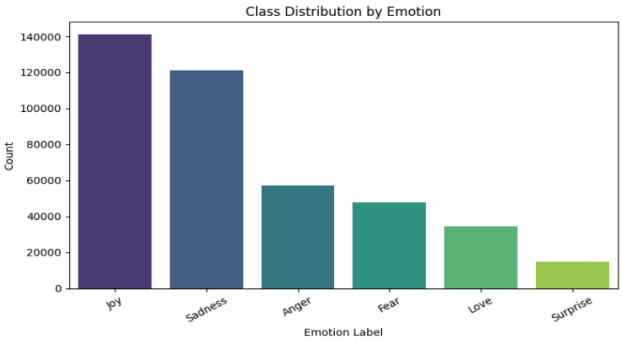

Figure 1. Class distribution by emotion label. Joy dominates at 33.8%, while Surprise is the smallest class at 3.6%.

Table 4. Target variable distribution — emotion class frequencies and modelling implications.

Code	Emotion	Count	%	Note
1	Joy	141,067	33.84	Largest class
0	Sadness	121,187	29.07	Second largest
3	Anger	57,317	13.75	Mid-tier
4	Fear	47,712	11.45	Overlap w/ Surprise
2	Love	34,554	8.29	Minority class
5	Surprise	14,872	3.59	Highest misclass. risk
	Total	**416,809**		IR $\approx 9.5:1$

3.2.1. Class Distribution Analysis

Figure 1 shows the absolute counts per class alongside the proportional breakdown. The distribution reveals a significant imbalance: Joy and Sadness together account for nearly 63% of all samples, while Surprise represents fewer than 4%. Table 4 presents the full class distribution with modelling implications.

3.2.2. Text Length and Word Count Distribution

Table 5 presents descriptive statistics for structural text features by emotion class. Class-wise comparisons reveal observable differences in average text length across emotions,

Table 5. Numerical summary statistics — `text_length` by emotion class.

Class	Mean	Med.	Std	Min	Max	Note
Overall	97.0	86	56.2	11	512	Right-skewed
Anger	96.0	84	57.1	11	512	Concise tone
Fear	96.7	85	56.3	12	512	Narrative
Joy	98.8	88	55.5	12	512	Celebratory
Love	104.7	94	56.4	13	512	Longest avg.
Sadness	93.1	81	53.9	11	512	Compressed
Surprise	99.7	89	56.0	14	512	Exclamatory

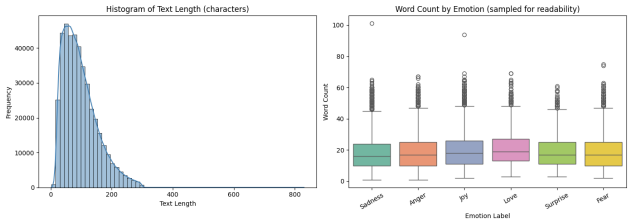

Figure 2. Histogram of text length distribution (left) and word count boxplot by emotion class (right).

suggesting that certain emotional expressions tend to be shorter (Anger, Sadness) or longer (Love, Surprise).

Figure 2 presents the histogram of text length distribution and word count boxplot by emotion class. The histogram confirms a right-skewed distribution with the majority of posts concentrated between 20 and 150 characters, and a long tail extending to 512. The boxplot reveals that the Love class has the highest median text length (median ≈ 94 chars), while Anger is the most concise (median ≈ 69 chars). These findings directly inform the maximum sequence length selection (512), truncation thresholds, and padding strategies for all neural network architectures.

3.2.3. Correlation Analysis

A Pearson correlation matrix (Figure 3) was computed to examine linear relationships among numerical features. A strong positive correlation ($r = 0.984$) was observed between `text_length` and `word_count`, confirming structural redundancy — only one need be retained as a feature. In contrast, the correlation between the numeric label encoding and length-based features is near-zero ($r < 0.03$), confirming that emotional classification depends primarily on semantic and contextual content rather than message length. This reinforces TF-IDF and embedding-based representations as the essential feature extraction strategy.

3.3. EDA Summary and Modelling Implications

Table 6 consolidates the key EDA dimensions with their findings, modelling implications, and required actions before model development.

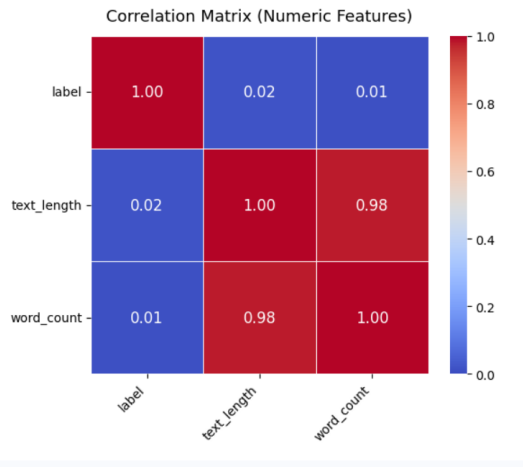

Figure 3. Pearson correlation heatmap among numeric features. The near-zero correlation between `label` and length features confirms that semantic content drives classification.

3.4. Proposed Architecture

The proposed methodology follows a structured NLP workflow: preprocessing $\rightarrow$ feature extraction $\rightarrow$ model development $\rightarrow$ evaluation $\rightarrow$ deployment. The full pipeline is illustrated in Figures 4 and 5.

Preprocessing. Text normalisation includes lowercasing, URL and special character removal, stopword elimination, and lemmatisation. Informal abbreviations and chat words are standardised to reduce lexical noise. Stopword removal uses the NLTK English stopword list; however, negation words and intensifiers (*e.g.*, "not", "never", "very", "extremely") are retained via a custom exception list, as these tokens carry direct emotional polarity and their removal would cause loss of discriminative signal. Class imbalance is addressed through stratified sampling and class-weighted learning, with SMOTE resampling considered as an alternative strategy.

Feature Representation. For machine learning models, TF-IDF vectorisation transforms text into numerical feature representations:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log\left(\frac{N}{\text{DF}(t)}\right) \tag{1}$$

where $\text{TF}(t, d)$ is the term frequency, $\text{DF}(t)$ is the document frequency, and N is the total number of documents. For deep learning models, an embedding layer converts tokens into dense vector representations.

Data Splitting. The dataset is split 80% training / 20% testing with a 20% validation holdout from training data, using stratified sampling to preserve class proportions.

Evaluation Metrics. Model performance is evaluated

Table 6. EDA summary — key findings, modelling implications, and required actions.

EDA Dimension	Key Finding	Modelling Implication	Required Action
Class distribution	Joy ≈33.8%; Surprise ≈3.6%	Use macro-F1 and weighted-F1	Class weights or SMOTE
Text length	Right-skewed; median ≈ 85 chars	Max seq. length = 512 for DL	Truncation and padding
Word count	$r \approx 0.99$ with text_length	Structurally redundant	Auxiliary feature only
Null values	0.00% null rate	No imputation required	Dataset ready
Emotion–length	Weak correlation ($r < 0.03$)	Semantic content is primary	TF-IDF & embeddings essential

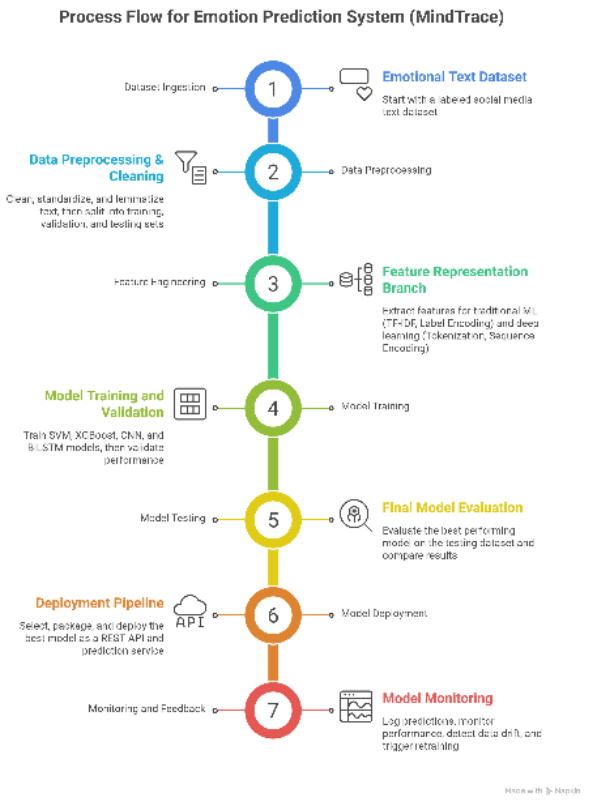

Figure 4. MindTrace process flow — from raw dataset through preprocessing, feature extraction, model training, evaluation, and deployment.

using:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

4. Proposed Algorithms

This study implements and compares four classification models selected to represent both traditional machine learning and deep learning approaches under identical experimental conditions.

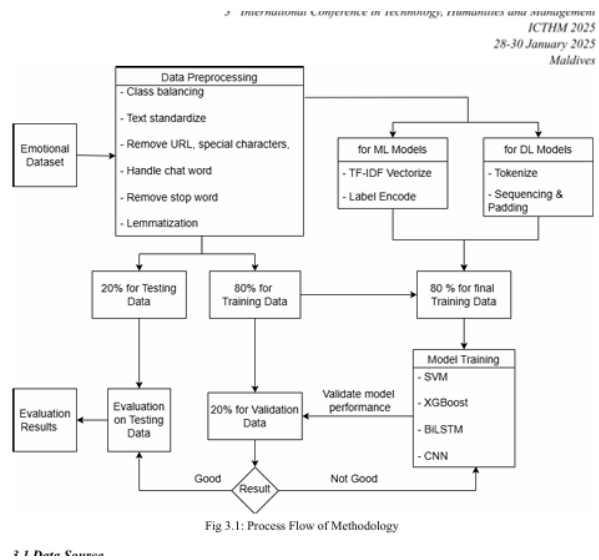

Figure 5. Detailed pipeline architecture showing preprocessing branches for ML (TF-IDF) and DL (tokenisation and padding) models, training, validation, and evaluation workflow.

4.1. Support Vector Machine (SVM)

SVM serves as a strong baseline classifier by maximising the margin between classes (Dzikovska et al., 2012). The optimisation objective is defined as:

$$\min_{w,b} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i \tag{4}$$

subject to $y_i(w \cdot x_i + b) \geq 1 - \xi_i$ and $\xi_i \geq 0$, where w is the weight vector, b is the bias term, C is the regularisation parameter, and ξ_i are slack variables permitting soft-margin classification.

4.2. Extreme Gradient Boosting (XGBoost)

XGBoost is applied as an ensemble learning method that minimises a regularised objective function (Chen and Guestrin, 2016):

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \tag{5}$$

where l is the loss function measuring prediction error, $\hat{y}_i$ is the predicted value, and $\Omega(f_k)$ is the regularisation term penalising model complexity.

4.3. Convolutional Neural Network (CNN)

CNN models extract local n-gram features through convolution and pooling layers (Hasan et al., 2019). The convolutional operation applies learned filters over contiguous word sequences, capturing phrase-level patterns that are informative for emotion classification. Max-pooling retains the most salient features from each filter map.

4.4. Bidirectional LSTM (BiLSTM)

BiLSTM captures bidirectional contextual dependencies in sequential text by processing token sequences in both forward and backward directions (Sun et al., 2023). This architecture is particularly suited for emotion classification where the meaning of a word may depend on both preceding and following context.

4.5. Evaluation Framework

All models are evaluated under identical experimental conditions. Both macro-averaged F1 (equal class weighting, sensitive to minority class performance) and weighted-averaged F1 (proportional to class size) are computed to address class imbalance. Confusion matrices provide class-wise predictive analysis, with particular focus on semantically similar emotion pairs (fear/surprise; sadness/love). The 80/20 stratified train-test split preserves class proportions across all subsets, ensuring consistent class representation throughout training and evaluation.

4.6. Deployment Architecture

Following evaluation, the best-performing model will be serialised and integrated into a lightweight REST API deployment pipeline. The selected model (along with its TF-IDF vectoriser or embedding weights) will be packaged using Flask or FastAPI and containerised with Docker to ensure reproducibility. The inference endpoint will accept raw text input, apply the same preprocessing pipeline used during training, and return a predicted emotion label with class probability scores. Future extensions may include a web-based dashboard for real-time emotion monitoring and a retraining pipeline triggered by incoming labelled data.

5. Preliminary Results

At the current proposal stage, formal model training has not yet been conducted. However, preliminary observations drawn from EDA provide a structured basis for expected performance benchmarks.

Expected baseline performance. Based on the literature (Dzikovska et al., 2012; Wang, 2018), SVM with TF-IDF is anticipated to achieve macro-F1 scores in the range of 0.70–0.78 on a balanced multi-class emotion dataset of this scale, serving as the primary classical baseline. XGBoost is expected to perform similarly or marginally below SVM due to the high dimensionality of TF-IDF features, which can limit tree-based ensemble effectiveness without dimensionality reduction.

Expected deep learning performance. CNN models are expected to outperform classical baselines due to their ability to capture local phrase-level patterns (Hasan et al., 2019). BiLSTM is anticipated to achieve the highest overall performance, with macro-F1 scores projected in the range of 0.80–0.87, owing to its bidirectional contextual encoding capability (Sun et al., 2023). However, BiLSTM performance is expected to be most sensitive to class imbalance, particularly for the Surprise class (3.59% of samples).

Expected misclassification patterns. EDA word frequency analysis (Figure 2) indicates significant vocabulary overlap between semantically adjacent emotion pairs — particularly fear/surprise and sadness/love. These pairs are expected to exhibit the highest inter-class confusion rates across all four models. This is consistent with prior findings in multi-class emotion classification literature (Nandwani and Verma, 2021; Seyeditabari et al., 2018).

Planned preliminary experiment. Prior to full-scale training, a preliminary experiment will be conducted using a stratified 10% subsample ($\approx$ 41,680 records) to validate the preprocessing pipeline, confirm label encoding integrity, and generate initial macro-F1 estimates for all four models.

6. Contribution and Expected Outcomes

6.1. Contribution of the Study

This study contributes to the field of emotion detection by providing a structured comparative evaluation of traditional machine learning and deep learning approaches under a unified experimental framework. Unlike studies that evaluate models under varying preprocessing conditions or heterogeneous datasets, this research implements a consistent NLP pipeline, standardised dataset partitioning strategy, and uniform evaluation metrics to ensure fair and interpretable comparison.

The study further contributes by analysing the influence of feature representation methods — including TF-IDF and embedding-based approaches — within the same dataset context. Through class-wise performance evaluation and confusion matrix analysis, the research provides deeper insight into model behaviour across discrete emotion categories. Concretely, this study will produce a reproducible experimental benchmark — including preprocessing scripts, model training code, and a performance comparison table — made available for future researchers to directly extend or compare against on the same 416,809-sample Twitter corpus.

6.2. Expected Outcomes

Deep learning models, particularly BiLSTM, are expected to outperform traditional ML algorithms due to their ability to

capture contextual and sequential dependencies. CNN models are anticipated to demonstrate competitive performance through effective local feature extraction, while SVM and XGBoost are expected to provide reliable baseline results with lower computational complexity. Certain emotion pairs — particularly fear/surprise and sadness/love — are expected to yield higher misclassification rates due to semantic similarity, providing meaningful insight into the limits of each architectural approach.

7. Conclusion and Future Work

This proposal has argued that the dominant limitation in current emotion classification research is not model capability but experimental inconsistency — studies differ in dataset, preprocessing, and evaluation metric, making performance comparisons unreliable. MindTrace addresses this directly by fixing all experimental conditions and varying only the model, isolating intrinsic differences between SVM, XGBoost, CNN, and BiLSTM on a 416,809-sample Twitter corpus.

The EDA presented in Section 3 confirms that the dataset is structurally sound, class-imbalanced at a ratio of approximately $9.5:1$, and that semantic overlap between adjacent emotion categories (fear/surprise, sadness/love) will be the primary challenge for all models. The architectural specifications in Section 4 ensure the study is fully reproducible.

If the comparative results confirm that BiLSTM outperforms classical baselines on minority emotion classes under macro-F1 — as the literature suggests — this will provide concrete, dataset-specific evidence that deep contextual architectures are worth the additional computational cost for emotion-aware applications in social media analytics, mental health monitoring, and human-computer interaction.

Future work will extend the framework in four directions:

1. Fine-tuning transformer-based models (BERT, RoBERTa) on the same dataset to compare against the baselines established here.
2. Cross-domain evaluation extending the pipeline to non-Twitter corpora such as Reddit or product reviews to assess generalisation.
3. Real-time deployment as a production REST API with a web-based emotion monitoring dashboard.
4. Explainability analysis using SHAP and LIME to identify the token-level features that most discriminate between semantically similar emotion classes.

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